Project

E-Commerce

Ecommerce Data Analysis to Increase Revenue & Customer Insights (SQL + Dashboard)

Tools: SQLite, DBeaver, Power BI

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Business Problem & Solution

Problem:
Many ecommerce businesses struggle with:
- Missing or incomplete customer data
- No visibility on top-performing products
- Unclear sales channel performance

Solution:
In this project, I analyzed 541K+ ecommerce transactions using SQL to:
- Identify key revenue drivers
- Detect data quality issues
- Segment sales channels for better decision-making

Result:
This analysis provides actionable insights for increasing revenue, improving customer targeting, and optimizing business strategy.

Data Summary Statistik

DBeaver 25.3.0 - <Ecommerce.db> ecommerce pemeriksaan data 02.sql

```sql
SELECT
    COUNT(*) AS total_rows,
    COUNT(DISTINCT InvoiceNo) AS unique_invoices,
    COUNT(DISTINCT CustomerID) AS unique_customers,
    COUNT(DISTINCT StockCode) AS unique_products,
    MIN(Quantity) AS min_qty,
    MAX(Quantity) AS max_qty,
    ROUND(AVG(Quantity), 2) AS avg_qty,
    MIN(UnitPrice) AS min_price,
    MAX(UnitPrice) AS max_price,
    ROUND(AVG(UnitPrice), 2) AS avg_price,
    SUM(Quantity * UnitPrice) AS total_revenue
FROM ecommerce;
```

Results 1

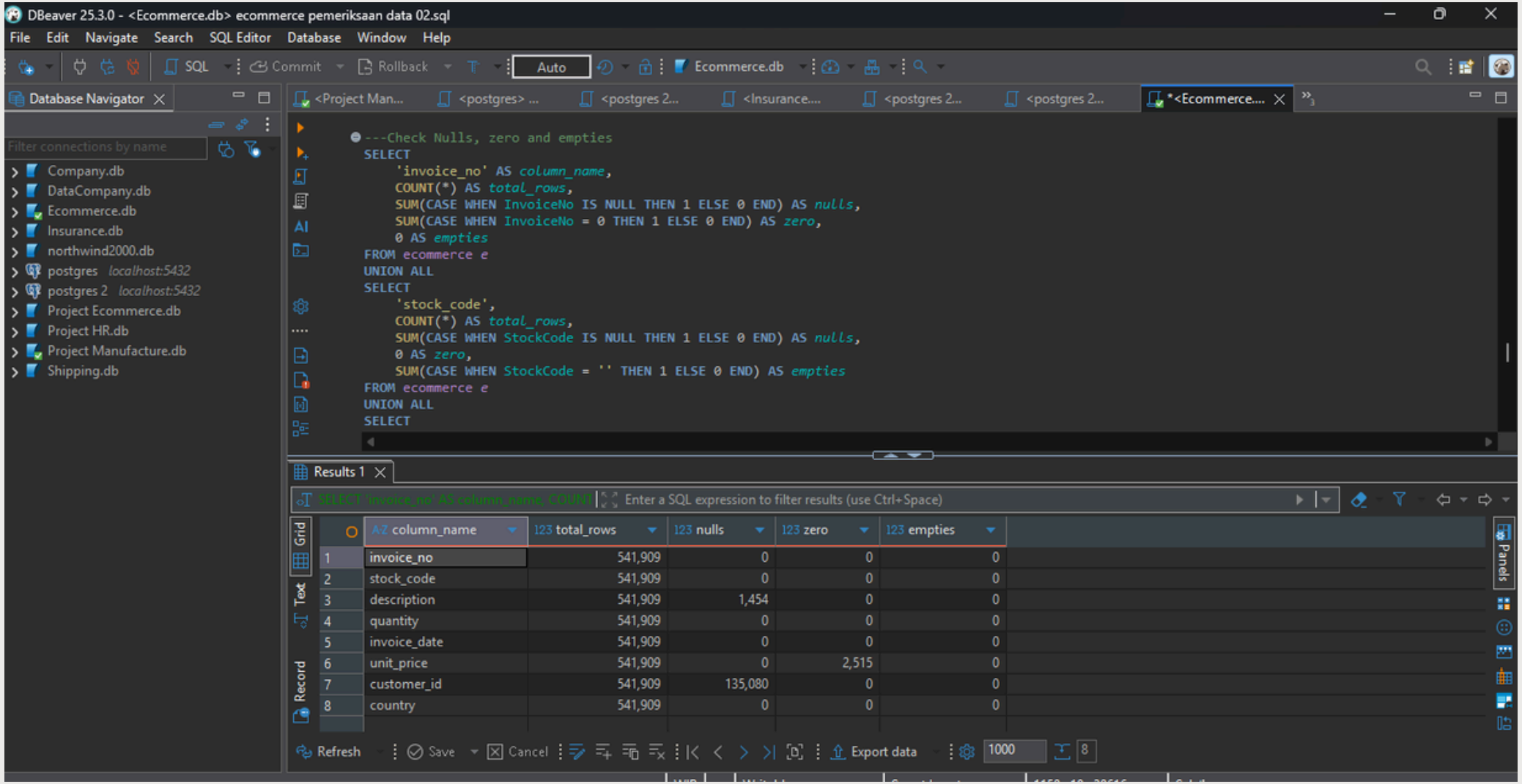
	123 total_rows	123 unique_invoices	123 unique_customers	123 unique_products	123 min_qty	123 max_qty	123 avg_qty	123 min_price	123 max_price	123 avg_price	123 total_revenue
1	541,909	25,900	4,372	4,070	-80,995	80,995	9.55	-11,062.06	38,970	4.61	9,747,747.934

- total_rows 541,909 transaction s
- unique_invoices 25,900
- unique_customers 4,372
- unique_products 4,070
- min_qty -80,995
- max_qty 80,995
- avg_qty 9.55
- min_price -11,062.06
- max_price 38,970
- avg_price 4.61
- total_revenue 9,747,747.934

Notes:

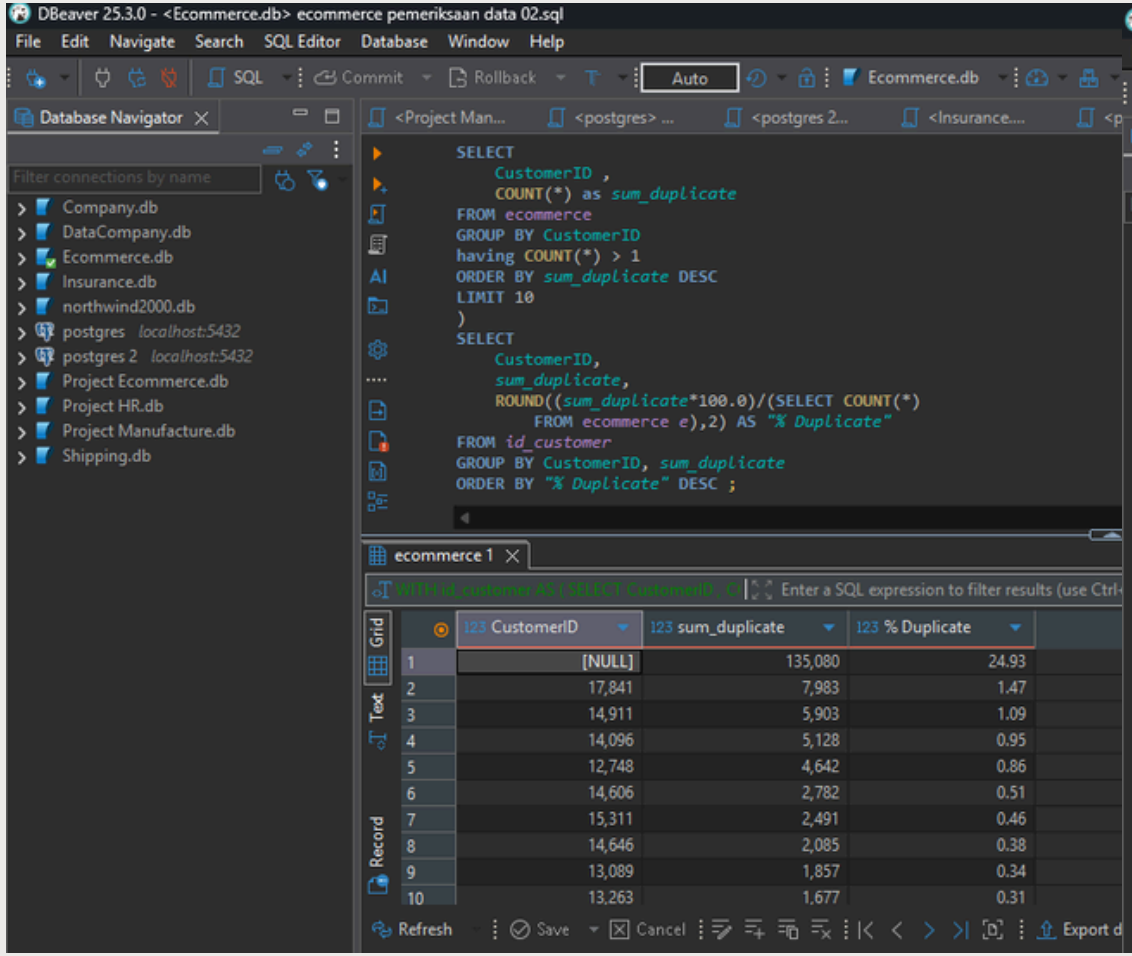
- Negative quantity indicates returns/refunds
- minus price maybe free items or data entry error

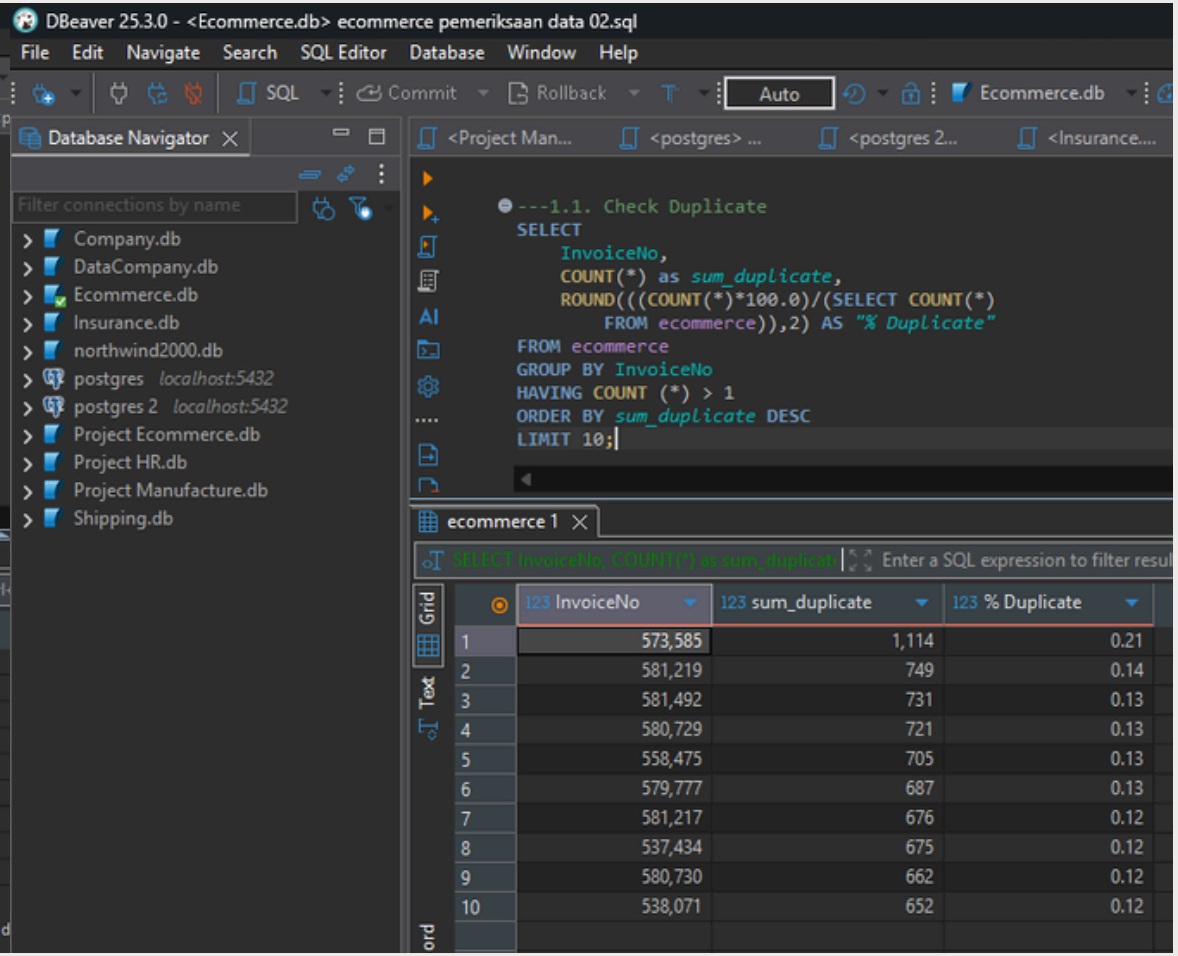
Data Quality Check (Null, Zero and Empty)



- Using Case - End, checking null, zero and empty row in column
- Customer_id has nulls 135k rows, it’s a big problem, it will influence customer segmentation / behavior
- Description has nulls 1,454, maybe no updates information from admin
- Unit_price has zero 2,515, maybe no updates information from admin

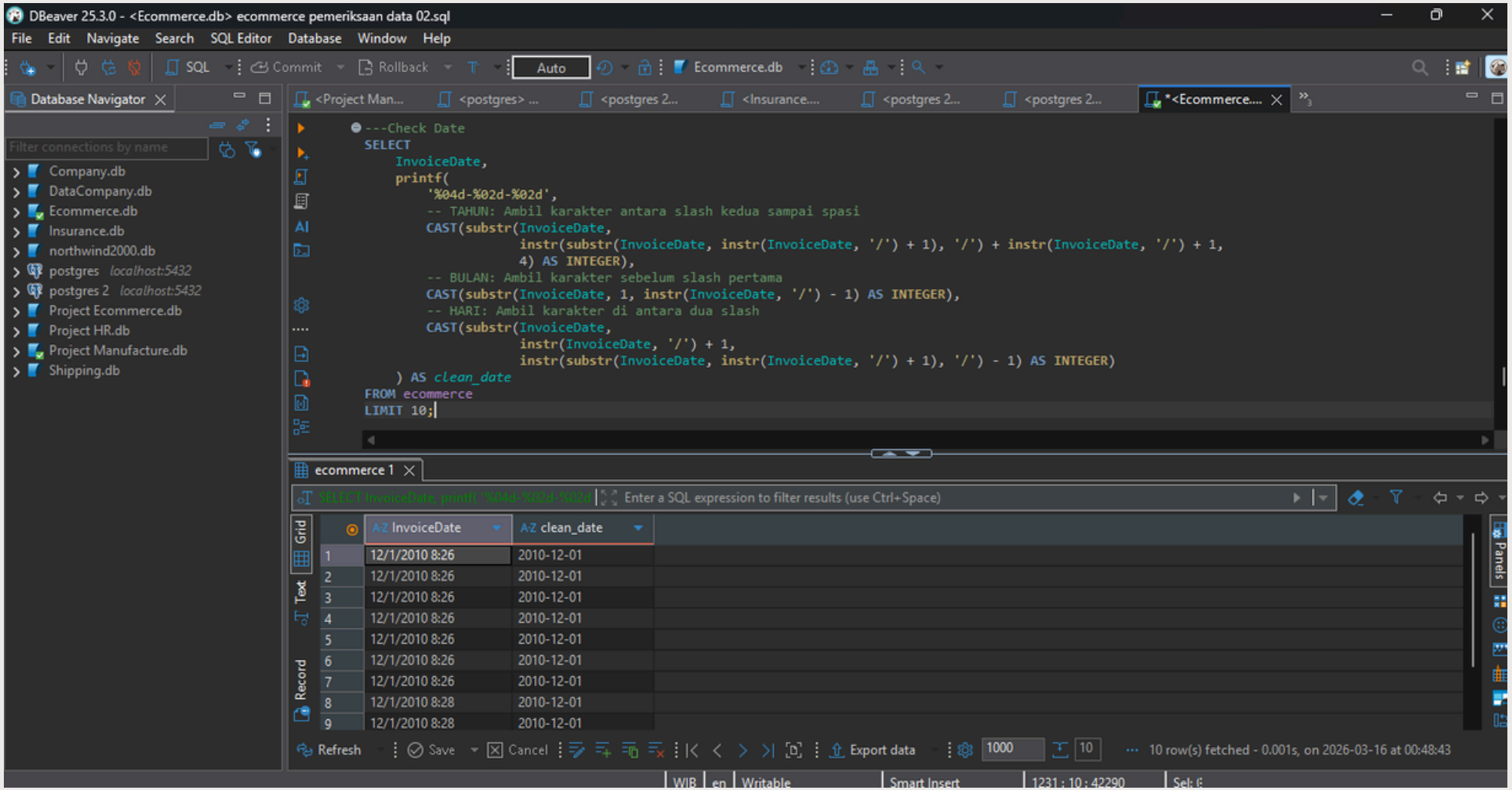
Data Duplicate Check





- Identified duplicate transactions in ecommerce dataset with 541,909 rows
- Found InvoiceNo 573585 appears 1114 times (0.21% of data). Duplicate transactions detected, which may lead to inaccurate revenue reporting and financial misinterpretation.
- 25% of transactions lack customer data → this reduces ability to do customer targeting and retention campaigns
- Used SQL aggregation and window functions for analysisgraph text

Data Cleaning Date Transformation



- Transformed inconsistent date format (MM/DD/YYYY HH:MM) to ISO standard (YYYY-MM-DD)
- Used SQL string functions: substr(), instr(), printf()
-

Before (transformation date)	After (transformation date)
12/1/2010 8:26	2010-12-01
12/1/2010 8:28	2010-12-01

Sales Channel Creation & Analysis

```sql
CREATE TABLE SalesChan AS
  SELECT *,
      CASE
          WHEN Country = 'Hong Kong' THEN 'RetailStore'
          WHEN Country = 'Unspecified' THEN 'GuestCheckout'
          WHEN CustomerID IS NULL OR CustomerID = '' THEN 'Anonymous'
              ELSE 'OnlineRegistered'
      END AS SalesChannel
  FROM ecommerce;

SELECT
    SalesChannel,
    COUNT(*) AS transaction_count,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM SalesChan), 2) AS percent,
    ROUND(SUM(Quantity * UnitPrice),2) AS total_revenue
FROM SalesChan
GROUP BY SalesChannel
ORDER BY total_revenue DESC;
```

SalesChan 1

	SalesChannel	transaction_count	percent	total_revenue
1	OnlineRegistered	406,585	75.03	8,297,398.74
2	Anonymous	134,590	24.84	1,435,482.36
3	RetailStore	288	0.05	10,117.04
4	GuestCheckout	446	0.08	4,749.79

4 row(s) fetched - 0.962s (0.006s fetch), on 2026-03-16 at 09:51:04

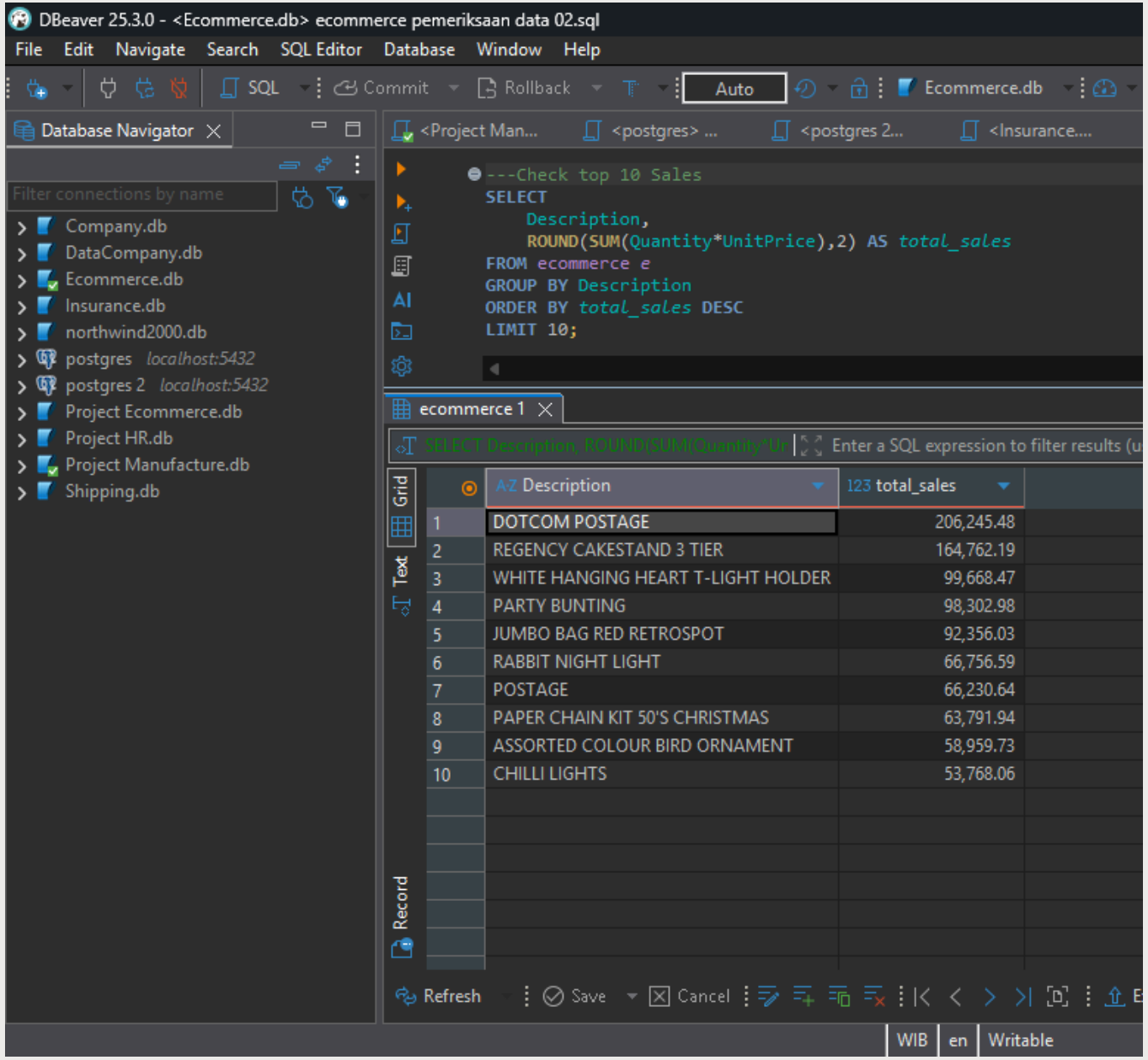
ASUMSI SALES CHANNEL:

- Hong Kong → RetailStore (physical store, no CustomerID needed)
- Unspecified → GuestCheckout (checkout without login)
- Other countries with CustomerID → OnlineRegistered
- Other countries with NULL CustomerID → Anonymous

ANALISIS:

- OnlineRegistered: 406,585 transaksi (75%, $8.3M revenue)
- Anonymous: 134,590 transaksi (24.8%, $1.44M revenue)
- RetailStore: 288 transaksi (0.05%, $10K revenue)
- GuestCheckout: 446 transaksi (0.08%, $4.7K revenue)

Top 10 Products



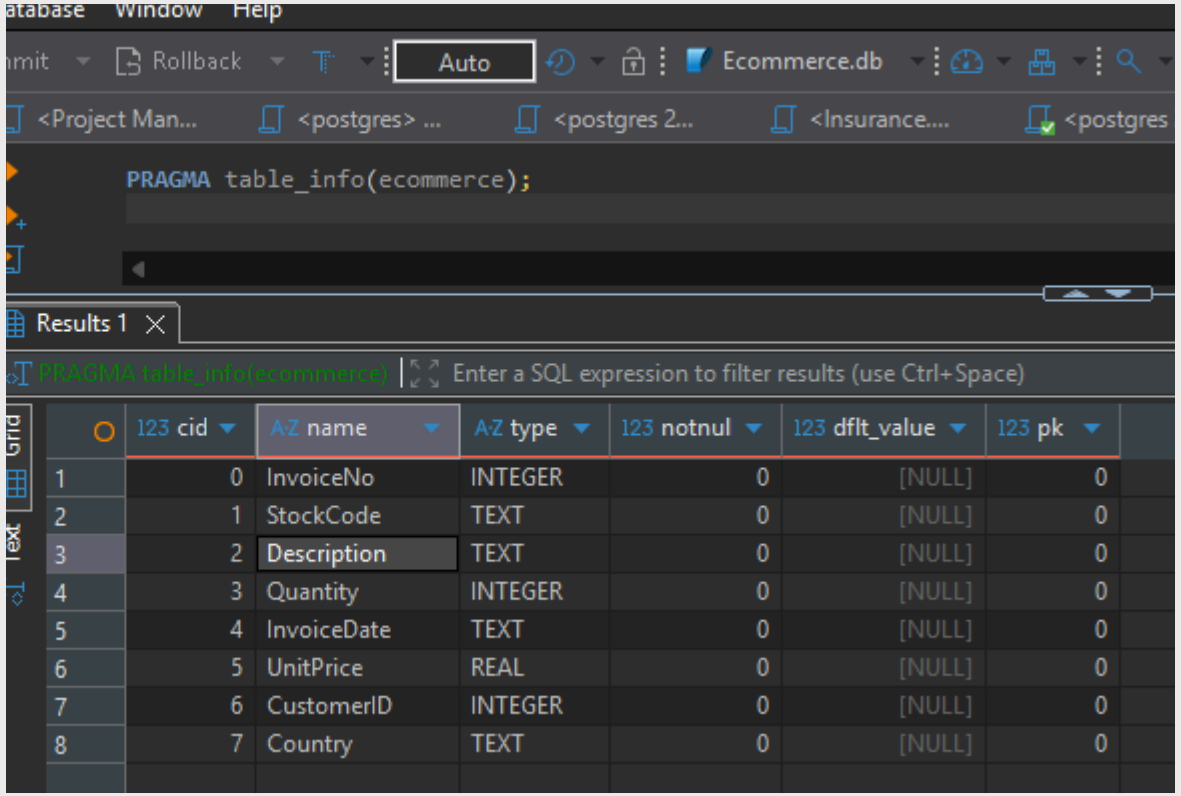
10 Best Selling Products

Description	total_sales
DOTCOM POSTAGE	206,245.48
REGENCY CAKESTAND 3 TIER	16,4762.19
WHITE HANGING HEART T-LIGHT HOLDER	99,668.47
PARTY BUNTING	98,302.98
JUMBO BAG RED RETROSPOT	92,356.03
RABBIT NIGHT LIGHT	66,756.59
POSTAGE	66,230.64
PAPER CHAIN KIT 50'S CHRISTMAS	63,791.94
ASSORTED COLOUR BIRD ORNAMENT	58,959.73
CHILLI LIGHTS	53,768.06

INSIGHT:

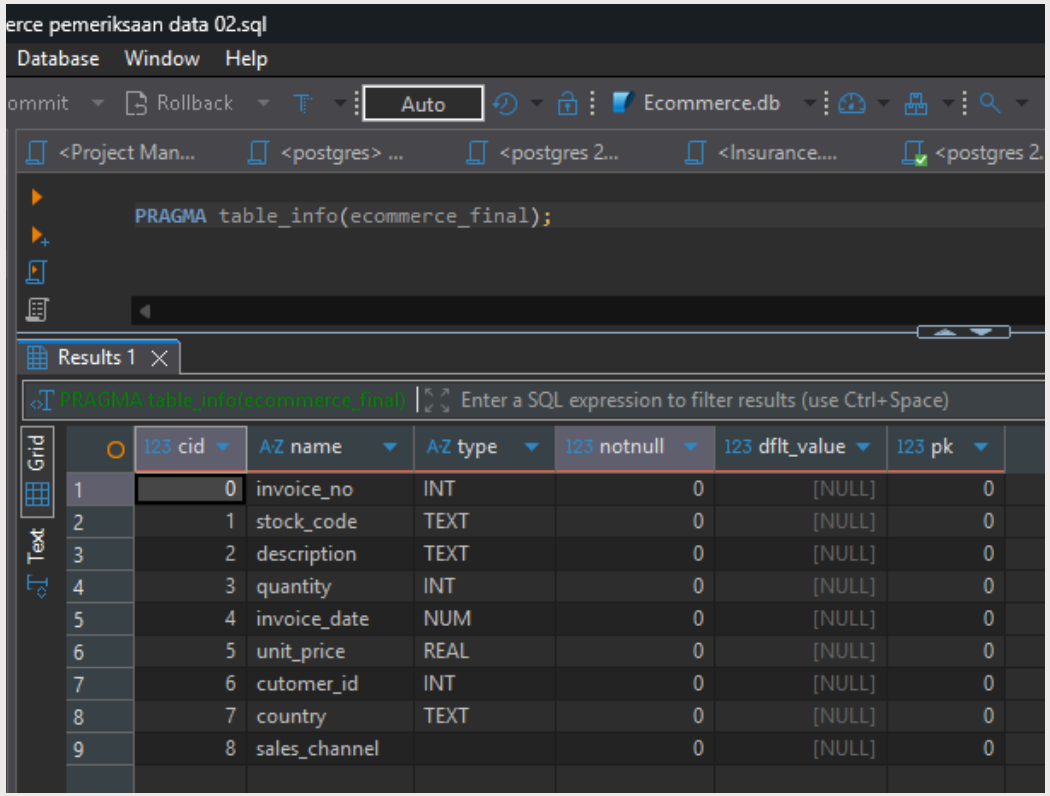
- Top 2 products contribute> $370,000 revenue

Final Table Structure



	cid	name	type	notnul	dflt_value	pk
1	0	InvoiceNo	INTEGER	0	[NULL]	0
2	1	StockCode	TEXT	0	[NULL]	0
3	2	Description	TEXT	0	[NULL]	0
4	3	Quantity	INTEGER	0	[NULL]	0
5	4	InvoiceDate	TEXT	0	[NULL]	0
6	5	UnitPrice	REAL	0	[NULL]	0
7	6	CustomerID	INTEGER	0	[NULL]	0
8	7	Country	TEXT	0	[NULL]	0

Before



	cid	name	type	notnull	dflt_value	pk
1	0	invoice_no	INT	0	[NULL]	0
2	1	stock_code	TEXT	0	[NULL]	0
3	2	description	TEXT	0	[NULL]	0
4	3	quantity	INT	0	[NULL]	0
5	4	invoice_date	NUM	0	[NULL]	0
6	5	unit_price	REAL	0	[NULL]	0
7	6	cutomer_id	INT	0	[NULL]	0
8	7	country	TEXT	0	[NULL]	0
9	8	sales_channel		0	[NULL]	0

After

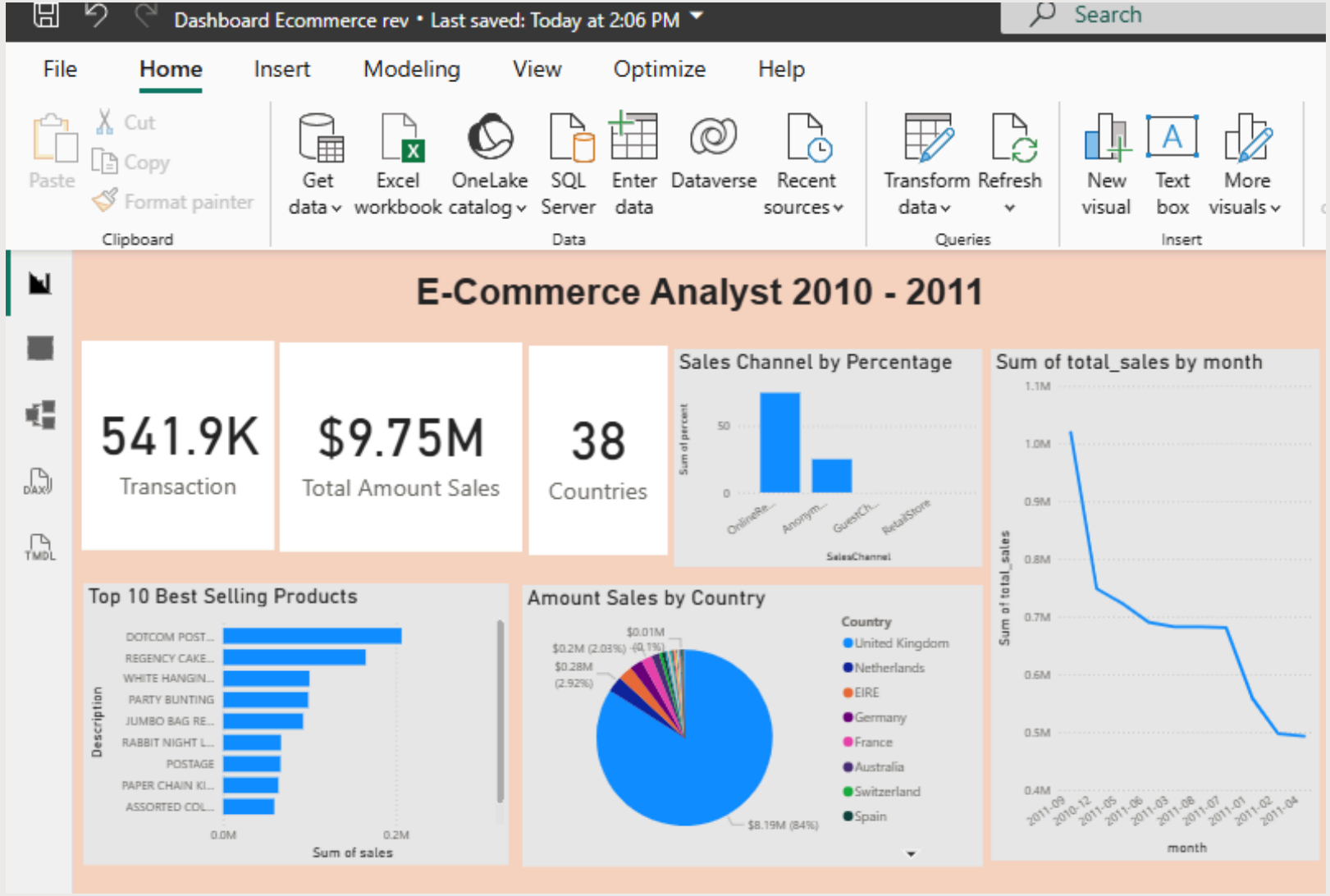
Changes :

- InvoiceDate → From TEXT to DATE (NUM - format ISO)
- Add column sales_channel for customer segmentation/behavior
- Root cause Data finished, continue for analysis: dashboard and forecasting.

Final Structure:

- invoice_no (INT)
- stock_code (TEXT)
- description (TEXT)
- quantity (INT)
- invoice_date (DATE/NUM)
- unit_price (REAL)
- customer_id (INT)
- country (TEXT)
- sales_channel (TEXT)

Dashboard by Power BI



DASHBOARD COMPONENTS:

- Total Revenue: $9.75M
- Top Country: UK (84%+ transactions)
- Sales by Channel: 75% OnlineRegistered
- Monthly Trend: Peak in September 2011
- Top Product: DOTCOM POSTAGE ($206K)

Summary

DATASET: 541,909 transaksi ecommerce (2010-2011)

DATA QUALITY ISSUES:
• 24.93% transaction without CustomerID (135,080 rows)
• 0.46% data with UnitPrice = 0 (2,515 rows)
• Inconsistent date structure

SOLUSTION:
✓ Cleaning date structure to standard ISO (YYYY-MM-DD)
✓ Changing column sales_channel for segmentation:
- OnlineRegistered (75% transactions, $8.3M revenue)
- Anonymous (24.8% transactions, $1.44M revenue)
- RetailStore & GuestCheckout (<0.2% transactions)
✓ Identification duplicate transactions
✓ Analyzing top products by revenue

KEY INSIGHTS:
1. Top 2 products have contribute > $370,000 revenue
2. 1 of 4 transaction didn't have customer data (opportunity for capture data)
3. Hong Kong transactions has 100% missing customer_id (validation: retail store)
4. UK has 27% missing customers (need more investigation)

DATASET READY FOR:
• Dashboard visualization (Power BI/Tableau)
• Time series analysis
• Customer segmentation
• Inventory planning

How I Can Help Your Business + CTA

Services I Offer:
- Data cleaning and preprocessing (SQL / Excel)
- Ecommerce sales analysis
- Customer segmentation
- Dashboard development (Power BI / Tableau)

Tools:
SQL, Excel, Power BI

I help businesses turn raw data into actionable insights to improve decision-making and increase revenue.

Let’s work together!

I’m available for freelance projects and ready to help you analyze your data for better business outcomes.